

Assignment 4 Report

Optimizing Transformer Translation with Ray Tune & Optuna

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March 10, 2026

1. Objective

This assignment optimizes a custom PyTorch Transformer for English-to-Hindi translation using Ray Tune with OptunaSearch and ASHA for early stopping to improve translation quality and training efficiency compared to a 100-epoch baseline.

2. Baseline Run (100 Epochs)

The baseline model was trained using the original fixed hyperparameters (no tuning).

Metric	Value
Total training epochs	100
Training time	111.53 minutes
Final training loss	0.0970
BLEU score (NLTK)	47.41 (0.4741)
Saved baseline model	<code>transformer_translation_final.pth</code>

Table 1: Baseline metrics from the notebook run.

Note: Baseline time was computed from notebook execution timestamps of the baseline training cell.

3. Ray Tune + Optuna Refactoring

The training loop was refactored into a configurable function `train_tune(config)` where model and optimizer hyperparameters are created from the Ray configuration per trial. Validation loss is reported to Ray Tune each epoch.

Hyperparameters Tuned and Search Ranges

At least 4 hyperparameters were required; 9 were tuned:

Search Configuration

- Search algorithm: **OptunaSearch**
- Scheduler: **ASHAScheduler** (`max_t=30`, `grace_period=5`, `reduction_factor=3`)
- Number of sampled trials: **12**

Hyperparameter	Search Range / Choices
Learning rate (<code>lr</code>)	<code>loguniform(1e-5, 1e-3)</code>
Batch size (<code>batch_size</code>)	<code>{16, 32, 64}</code>
Attention heads (<code>num_heads</code>)	<code>{4, 8}</code>
Feedforward dim (<code>d_ff</code>)	<code>{1024, 2048}</code>
Dropout (<code>dropout</code>)	<code>uniform(0.1, 0.4)</code>
Encoder/Decoder layers (<code>num_layers</code>)	<code>{4, 6}</code>
Model dim (<code>d_model</code>)	<code>{512}</code>
Gradient clipping (<code>grad_clip</code>)	<code>{1.0}</code>
Epochs per trial (<code>epochs</code>)	<code>{20, 25, 30}</code>

Table 2: Ray Tune search space used with OptunaSearch.

- Optimization objective: minimize validation loss (`metric=loss, mode=min`)
- GPU usage: 1 GPU per trial (single concurrent GPU trial)

4. Best Configuration Found

Parameter	Best Value
<code>lr</code>	2.95246×10^{-5}
<code>batch_size</code>	64
<code>num_heads</code>	4
<code>d_ff</code>	2048
<code>dropout</code>	0.107334
<code>num_layers</code>	6
<code>d_model</code>	512
<code>grad_clip</code>	1.0
<code>epochs</code>	30
Best validation loss from tuning	2.6154

Table 3: Best trial configuration selected by Ray Tune + Optuna.

5. Final Tuned Model Metrics

Using the best configuration, the model was retrained and evaluated:

6. Baseline vs Tuned Comparison

Efficiency Goal Status: Achieved. The tuned setup exceeded the target BLEU > 0.50 and beat baseline BLEU while using 30 epochs (significantly fewer than 100 epochs).

Metric	Value
Tuning sweep time	136.45 minutes
Final retraining time (best config)	32.28 minutes
Final retraining loss (epoch 30)	1.3345
BLEU score (NLTK)	64.99 (0.6499)
Saved tuned model	M25CSA010_ass_4_best_model.pth

Table 4: Final tuned model performance.

Metric	Baseline	Tuned
Epochs	100	30
BLEU (0–1 scale)	0.4741	0.6499
BLEU (percentage)	47.41	64.99
Final loss	0.0970	1.3345

Table 5: Main quality/efficiency comparison.

7. Conclusion

Ray Tune + Optuna successfully found a stronger hyperparameter configuration for the Transformer translation model. The final tuned model achieved substantial BLEU improvement (0.6499 vs 0.4741) with a much smaller epoch budget (30 vs 100), satisfying the assignment objective of better quality with improved training efficiency.

8. GitHub Repository

<https://github.com/Jyoti-Dwivedi-010/MLOps-Jyoti-Dwivedi-M25CSA010/tree/Assignment-4>